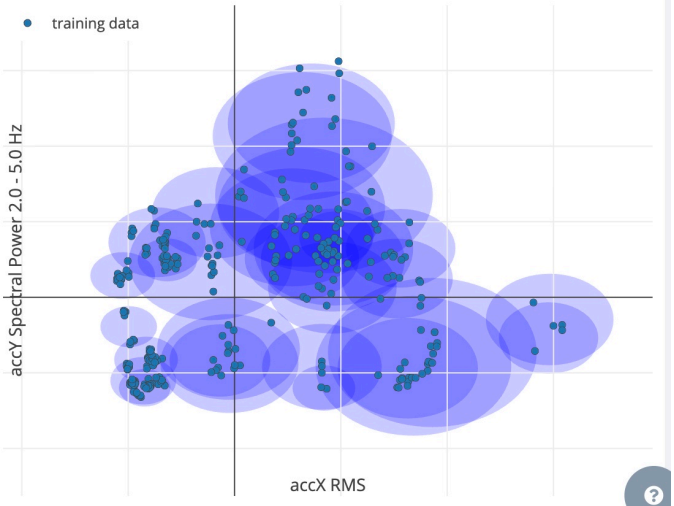
- [x] accX RMS
- [ ] accX Peak 1 Freq
- [x] accX Peak 1 Height
- [ ] accX Peak 2 Freq
- [ ] accX Peak 2 Height
- [ ] accX Peak 3 Freq
- [ ] accX Peak 3 Height
- [ ] accX Spectral Power 0.1 - 0.5 Hz
- [ ] accX Spectral Power 0.5 - 1.0 Hz
- [ ] accX Spectral Power 1.0 - 2.0 Hz
- [ ] accX Spectral Power 2.0 - 5.0 Hz
- [ ] accY RMS
- [ ] accY Peak 1 Freq
- [ ] accY Peak 1 Height
- [ ] accY Peak 2 Freq
- [ ] accY Peak 2 Height
- [ ] accY Peak 3 Freq
- [ ] accY Peak 3 Height
- [ ] accY Spectral Power 0.1 - 0.5 Hz
- [ ] accY Spectral Power 0.5 - 1.0 Hz
- [ ] accY Spectral Power 1.0 - 2.0 Hz
- [x] accY Spectral Power 2.0 - 5.0 Hz
- [ ] accZ RMS
- [ ] accZ Peak 1 Freq
- [ ] accZ Peak 1 Height
- [ ] accZ Peak 2 Freq
- [ ] accZ Peak 2 Height
- [ ] accZ Peak 3 Freq
- [ ] accZ Peak 3 Height
- [ ] accZ Spectral Power 0.1 - 0.5 Hz
- [ ] accZ Spectral Power 0.5 - 1.0 Hz
- [ ] accZ Spectral Power 1.0 - 2.0 Hz
- [x] accZ Spectral Power 2.0 - 5.0 Hz

```
Job completed (success)
```

Anomaly explorer

X Axis	Y Axis	Test data
accX RMS	accY Spectral Power 2	-- No test data



Lab 5 Part B Submission

Classification Reports

```
Epoch 1/100
41/41 [==============================] - 0s 1ms/step - loss: 7.5708
Epoch 2/100
41/41 [==============================] - 0s 885us/step - loss: 0.5628
Epoch 3/100
41/41 [==============================] - 0s 952us/step - loss: 0.2237
Epoch 4/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1787
Epoch 5/100
41/41 [==============================] - 0s 2ms/step - loss: 0.1515
Epoch 6/100
41/41 [==============================] - 0s 2ms/step - loss: 0.1477
Epoch 7/100
41/41 [==============================] - 0s 4ms/step - loss: 0.1447
Epoch 8/100
41/41 [==============================] - 0s 2ms/step - loss: 0.1429
Epoch 9/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1418
Epoch 10/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1389
Epoch 11/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1355
Epoch 12/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1324
Epoch 13/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1311
Epoch 14/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1257
Epoch 15/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1222
Epoch 16/100
41/41 [==============================] - 0s 3ms/step - loss: 0.1203
Epoch 17/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1170
Epoch 18/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1141
Epoch 19/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1157
Epoch 20/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1135
Epoch 21/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1087
Epoch 22/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1067
Epoch 23/100
41/41 [==============================] - 0s 994us/step - loss: 0.1055
Epoch 24/100
```

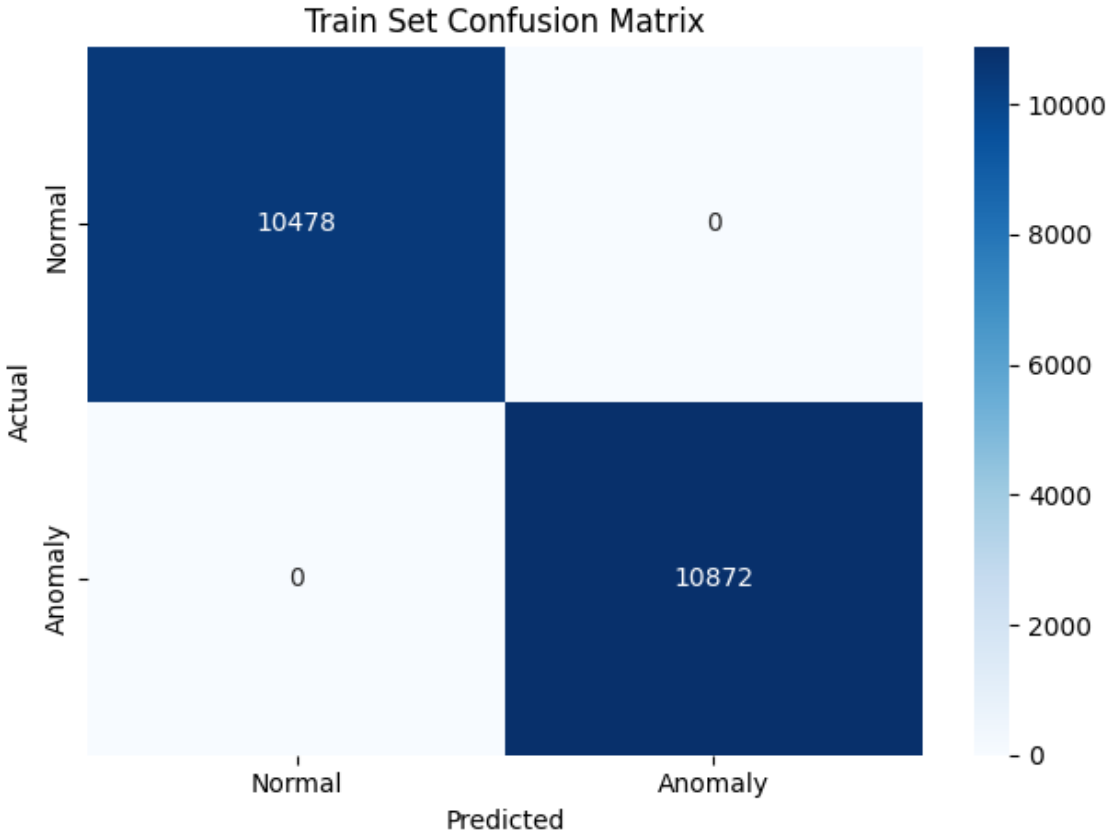
```
41/41 [==============================] - 0s 933us/step - loss: 0.1041
Epoch 25/100
41/41 [==============================] - 0s 924us/step - loss: 0.1037
Epoch 26/100
41/41 [==============================] - 0s 919us/step - loss: 0.1027
Epoch 27/100
41/41 [==============================] - 0s 930us/step - loss: 0.1028
Epoch 28/100
41/41 [==============================] - 0s 929us/step - loss: 0.1007
Epoch 29/100
41/41 [==============================] - 0s 951us/step - loss: 0.0997
Epoch 30/100
41/41 [==============================] - 0s 956us/step - loss: 0.1004
Epoch 31/100
41/41 [==============================] - 0s 1ms/step - loss: 0.1031
Epoch 32/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0974
Epoch 33/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0975
Epoch 34/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0974
Epoch 35/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0965
Epoch 36/100
41/41 [==============================] - 0s 976us/step - loss: 0.0954
Epoch 37/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0958
Epoch 38/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0953
Epoch 39/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0965
Epoch 40/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0950
Epoch 41/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0945
Epoch 42/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0951
Epoch 43/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0935
Epoch 44/100
41/41 [==============================] - 0s 953us/step - loss: 0.0968
Epoch 45/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0927
Epoch 46/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0917
Epoch 47/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0920
Epoch 48/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0947
Epoch 49/100
41/41 [==============================] - 0s 1000us/step - loss: 0.0908
Epoch 50/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0911
```

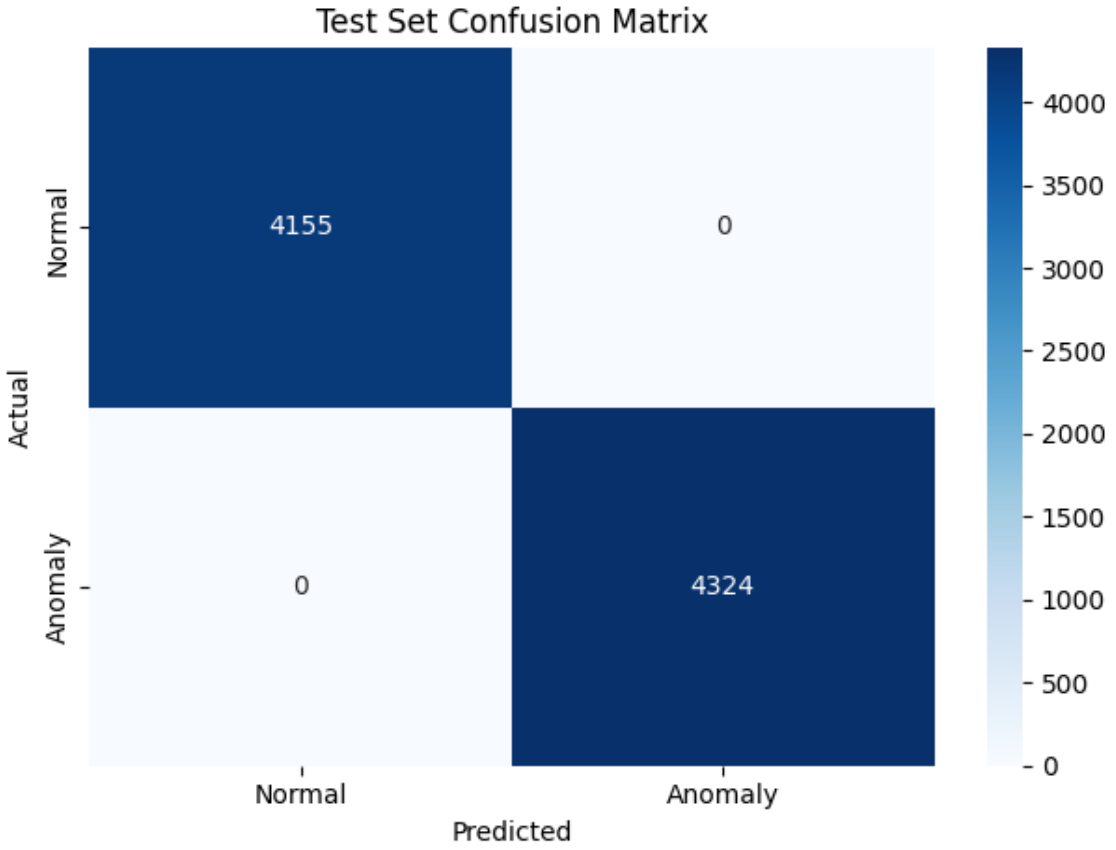
```
Epoch 51/100
41/41 [==============================] - 0s 973us/step - loss: 0.0914
Epoch 52/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0990
Epoch 53/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0913
Epoch 54/100
41/41 [==============================] - 0s 983us/step - loss: 0.0894
Epoch 55/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0904
Epoch 56/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0904
Epoch 57/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0931
Epoch 58/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0911
Epoch 59/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0896
Epoch 60/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0895
Epoch 61/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0884
Epoch 62/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0897
Epoch 63/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0901
Epoch 64/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0884
Epoch 65/100
41/41 [==============================] - 0s 997us/step - loss: 0.0891
Epoch 66/100
41/41 [==============================] - 0s 958us/step - loss: 0.0938
Epoch 67/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0871
Epoch 68/100
41/41 [==============================] - 0s 900us/step - loss: 0.0909
Epoch 69/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0873
Epoch 70/100
41/41 [==============================] - 0s 2ms/step - loss: 0.0911
Epoch 71/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0872
Epoch 72/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0866
Epoch 73/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0866
Epoch 74/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0904
Epoch 75/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0877
Epoch 76/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0857
Epoch 77/100
```

```
41/41 [==============================] - 0s 1ms/step - loss: 0.0908
Epoch 78/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0861
Epoch 79/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0882
Epoch 80/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0872
Epoch 81/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0874
Epoch 82/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0863
Epoch 83/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0856
Epoch 84/100
41/41 [==============================] - 0s 1ms/step - loss: 0.0878
Epoch 85/100
41/41 [==============================] - 0s 970us/step - loss: 0.0879
Epoch 86/100
41/41 [==============================] - 0s 917us/step - loss: 0.0851
Epoch 87/100
41/41 [==============================] - 0s 891us/step - loss: 0.0863
Epoch 88/100
41/41 [==============================] - 0s 898us/step - loss: 0.0858
Epoch 89/100
41/41 [==============================] - 0s 913us/step - loss: 0.0856
Epoch 90/100
41/41 [==============================] - 0s 869us/step - loss: 0.0941
Epoch 91/100
41/41 [==============================] - 0s 899us/step - loss: 0.0843
Epoch 92/100
41/41 [==============================] - 0s 846us/step - loss: 0.0850
Epoch 93/100
41/41 [==============================] - 0s 866us/step - loss: 0.0845
Epoch 94/100
41/41 [==============================] - 0s 851us/step - loss: 0.0856
Epoch 95/100
41/41 [==============================] - 0s 870us/step - loss: 0.0850
Epoch 96/100
41/41 [==============================] - 0s 884us/step - loss: 0.0862
Epoch 97/100
41/41 [==============================] - 0s 880us/step - loss: 0.0872
Epoch 98/100
41/41 [==============================] - 0s 876us/step - loss: 0.0848
Epoch 99/100
41/41 [==============================] - 0s 861us/step - loss: 0.0847
Epoch 100/100
41/41 [==============================] - 0s 862us/step - loss: 0.0843
328/328 [==============================] - 0s 283us/step
340/340 [==============================] - 0s 268us/step
130/130 [==============================] - 0s 281us/step
136/136 [==============================] - 0s 283us/step
Train Normal (label 0): 0.095492 ± 0.001413; Max Error: 0.943755; Min Error:
0.009353
```

```
Train Anomalous   (label 1): 37.978320 ± 0.335170; Max Error: 116.241899; Min Error: 2.108390
Test Normal         (label 0): 0.118477 ± 0.003029; Max Error: 1.209892; Min Error: 0.009989
Test Anomalous    (label 1): 38.446296 ± 0.567635; Max Error: 116.548179; Min Error: 1.922844
```

Confusion Matrices

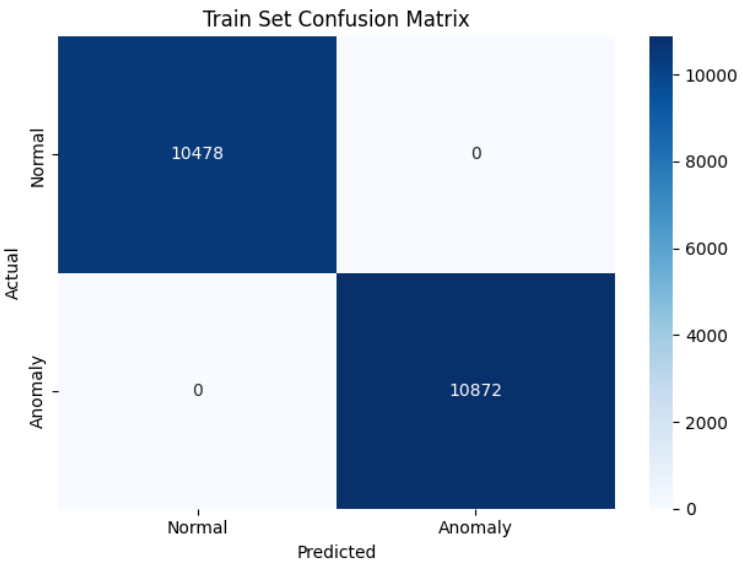


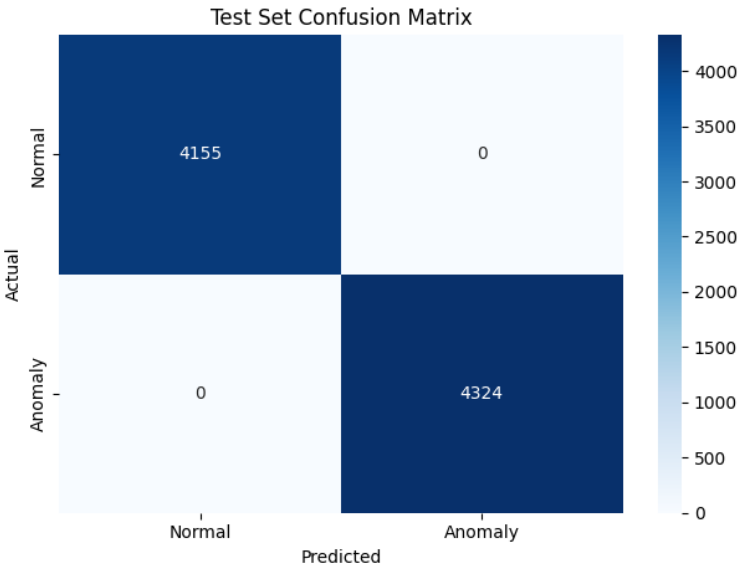


Classification Reports

```
668/668 [==============================] - 0s 300us/step
265/265 [==============================] - 0s 284us/step
 Train Set Evaluation
              precision    recall  f1-score   support

  Normal (0)       1.00      1.00      1.00     10478
 Anomaly (1)       1.00      1.00      1.00     10872

    accuracy                           1.00     21350
   macro avg       1.00      1.00      1.00     21350
weighted avg       1.00      1.00      1.00     21350


 Test Set Evaluation
              precision    recall  f1-score   support

  Normal (0)       1.00      1.00      1.00      4155
 Anomaly (1)       1.00      1.00      1.00      4324

    accuracy                           1.00      8479
   macro avg       1.00      1.00      1.00      8479
weighted avg       1.00      1.00      1.00      8479
```

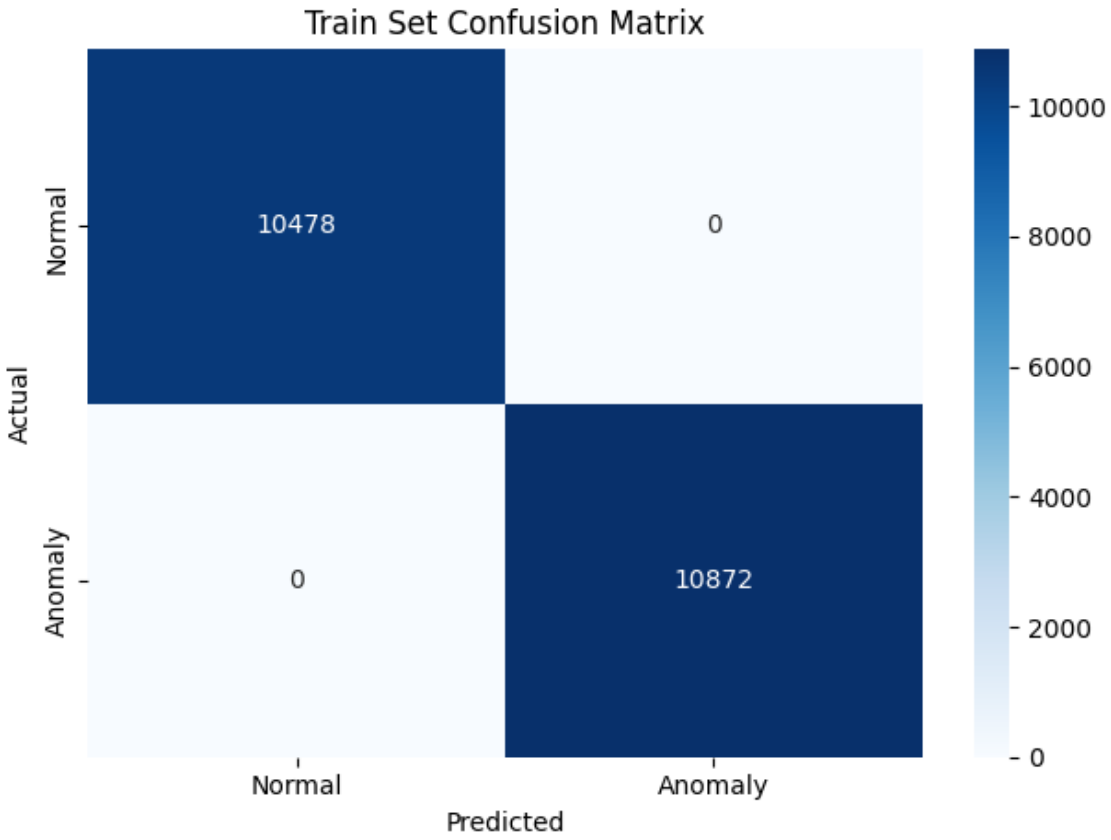
Confusion Matrices

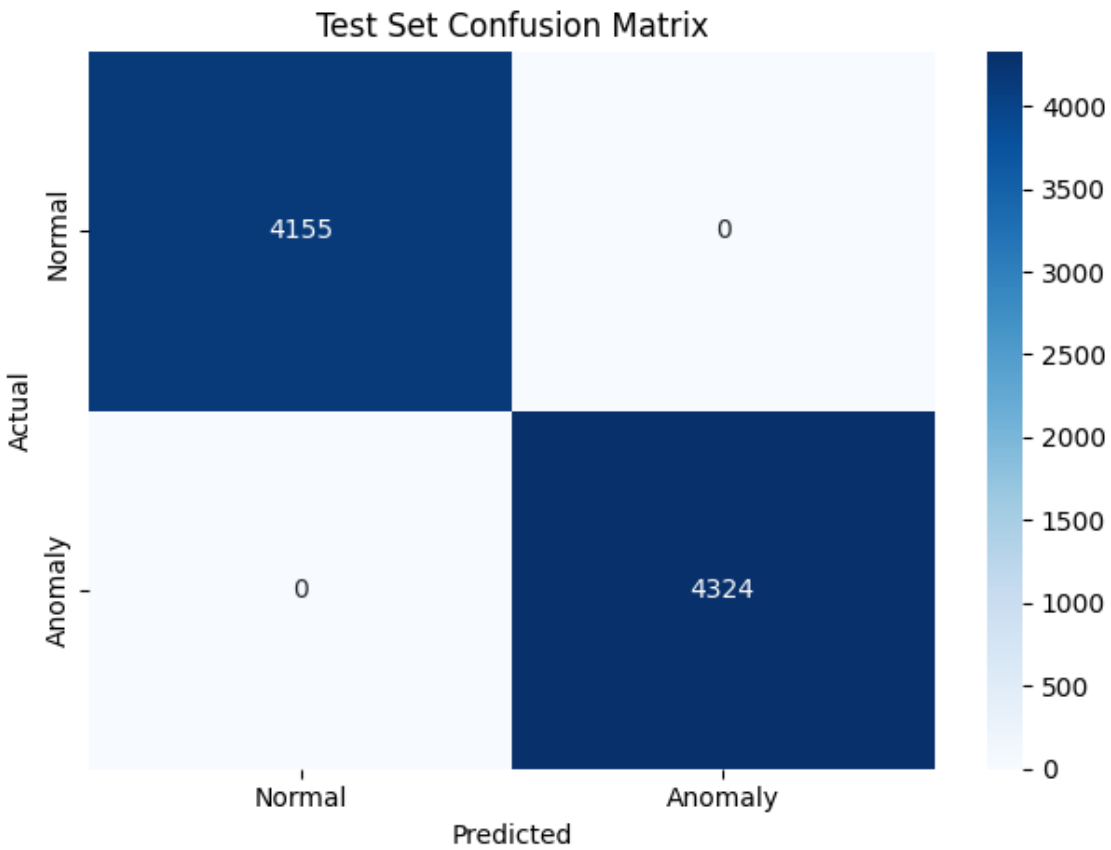




Classification Reports

Confusion Matrices



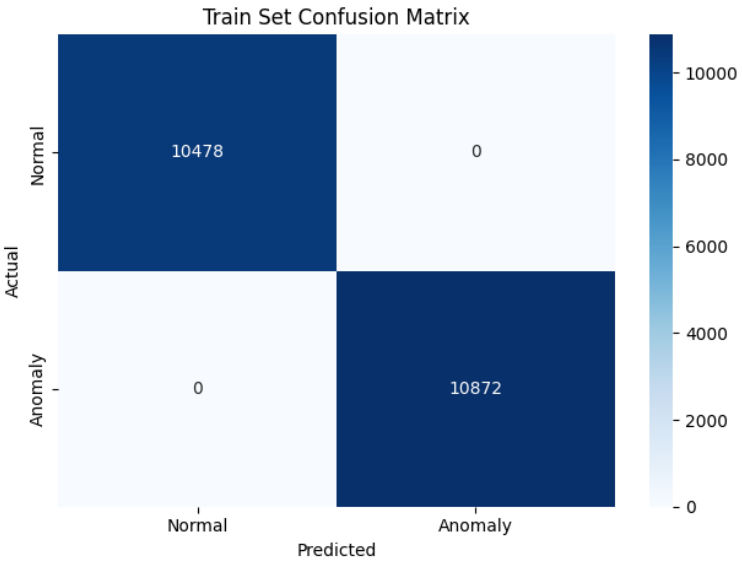


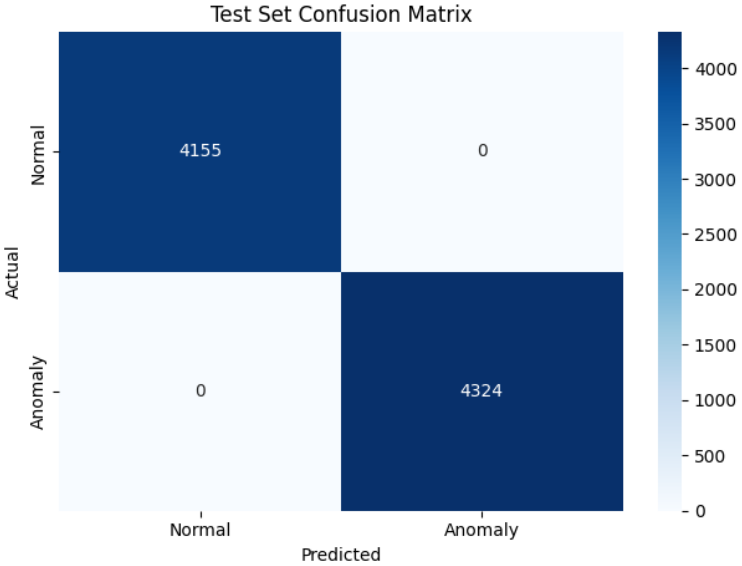
Classification Reports

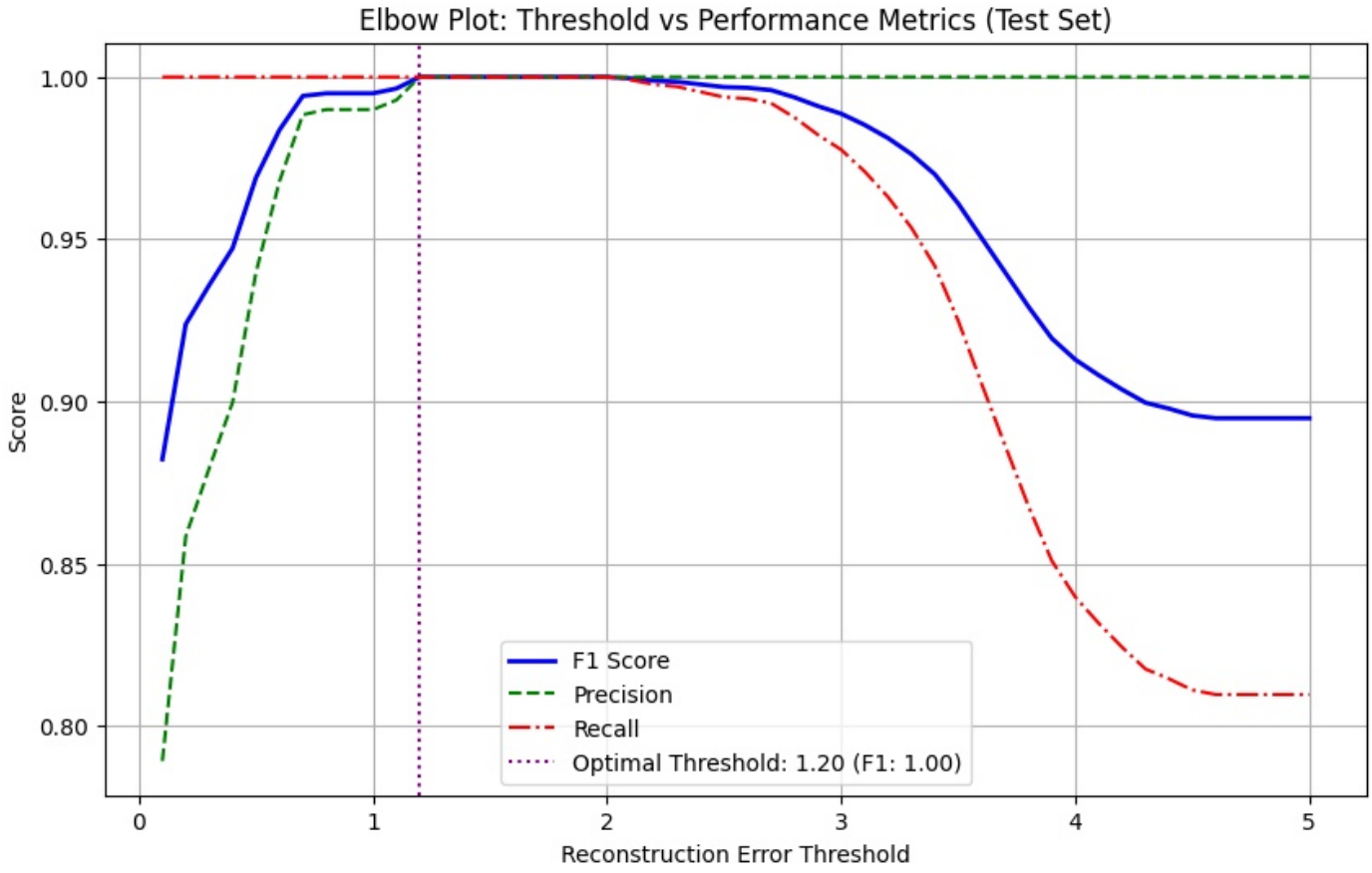
```
Train Set Evaluation
              precision    recall  f1-score   support
```

```
   Normal (0)       1.00      1.00      1.00     10478
  Anomaly (1)       1.00      1.00      1.00     10872

     accuracy                           1.00     21350
    macro avg       1.00      1.00      1.00     21350
 weighted avg       1.00      1.00      1.00     21350


 Test Set Evaluation
              precision    recall  f1-score   support

   Normal (0)       1.00      1.00      1.00      4155
  Anomaly (1)       1.00      1.00      1.00      4324

     accuracy                           1.00      8479
    macro avg       1.00      1.00      1.00      8479
 weighted avg       1.00      1.00      1.00      8479
```

Confusion Matrices




Test Set Confusion Matrix
Actual
Normal
Anomaly
4155
0
0
4324
Normal
Anomaly
Predicted
4000
3500
3000
2500
2000
1500
1000
500
0


Elbow Plot: Threshold vs Performance Metrics (Test Set)
Score
Reconstruction Error Threshold
1.00
0.95
0.90
0.85
0.80
0
1
2
3
4
5
F1 Score
Precision
Recall
Optimal Threshold: 1.20 (F1: 1.00)

```
18  const int kWindowSize =
```

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Na

```
Reconstruction error: 3.914453
🚨 Anomaly detected!
Reconstruction error: 3.979194
🚨 Anomaly detected!
Reconstruction error: 4.171315
🚨 Anomaly detected!
Reconstruction error: 4.070281
🚨 Anomaly detected!
```